

SMART INDIA HACKATHON 2026



TITLE PAGE

Problem Statement ID — SIH26129

Problem Statement Title — System integration and interoperability among government digital platforms, resulting in fragmented service delivery

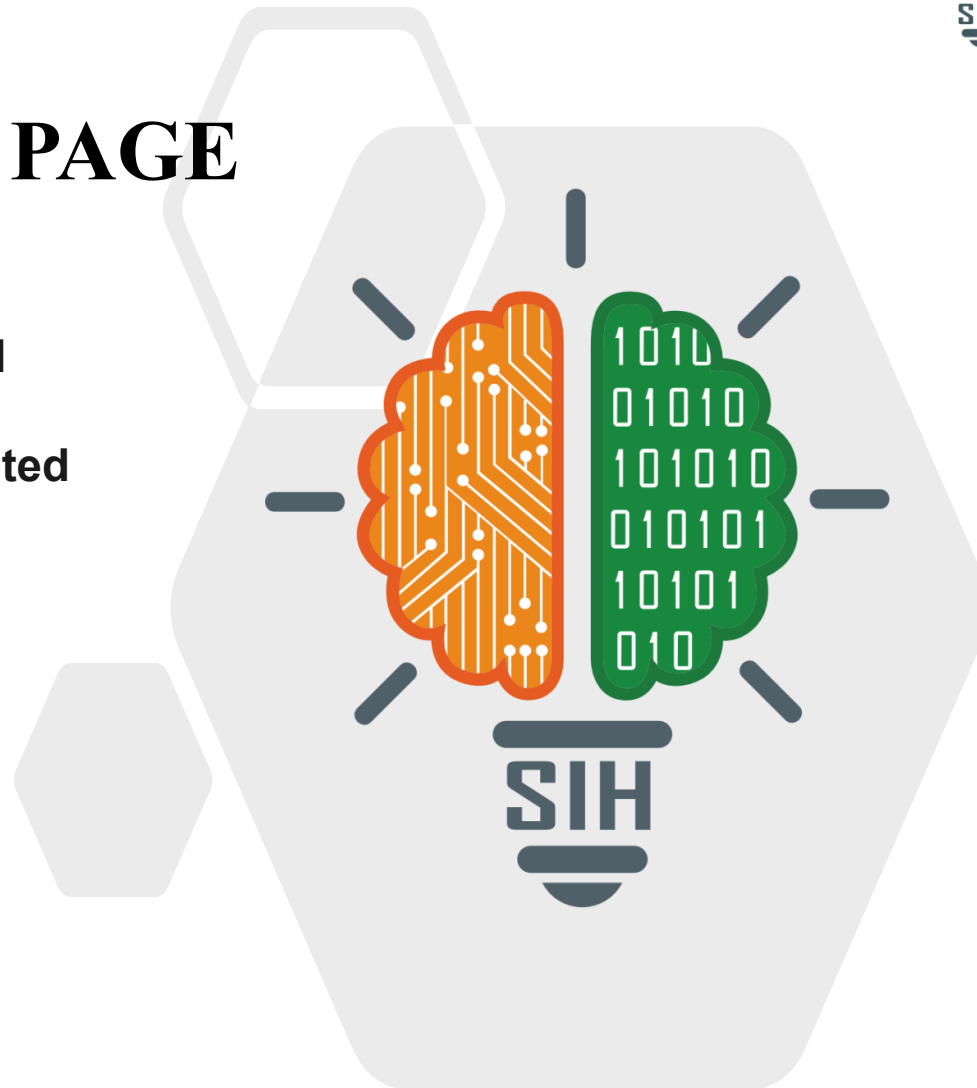
Theme — Miscellaneous

Organisation — Government of Maharashtra

PS Category — Software

Team ID — (fill from SIH portal after nomination)

Team Name (Registered on portal) — Samanvay



Not a new portal

Sits between existing Maharashtra systems. Does not replace Aaple Sarkar, Mahabhulekh, MahaDBT, Fire e-approval or MPCB/MAITRI.

Does not store papers

Certificates stay with the issuer. Samanvay owns links, consent, catalog, pointers, workflow, tracking, hash-chained audit.

Generic by construction

Scholarship, licence/NOC and farmer subsidy are three callers. Journey 3 is catalog configuration, not a new Java module.

How it addresses the problem

- Citizen facts (income, caste, 7/12, marks, fire NOC) are issued once by the department of record — then reused under consent.
- Connectors speak the department's protocol (REST / SOAP / SFTP / JDBC). DigiLocker is a consent pipe for issued copies, not the source of truth.
- One tracking reference across departments; officer desk retries when a source is down; audit proves what was asked, never a stored payload.

Innovation / uniqueness

- Control plane decides, data plane moves — no fetch without a 60-second, single-use access grant (typed, not a convention).
- Federated by default: pointers in the registry, live fetch, provenance on every field (“₹1,85,000 — Tahsildar Haveli”).
- Machines propose, humans dispose: probabilistic identity matches never auto-link. Build-time Modulith/ArchUnit fail illegal module edges.

Stack · Java 21 / Spring Boot 4 / Modulith · Postgres 16 · BPMN 2.0 · Keycloak · Resilience4j · REST · SOAP · SFTP · JDBC

Scholarship

Revenue · Education · DBT

Licence / NOC

Municipal · Fire · MPCB

Farmer subsidy

Land · Agri · DBT

SAMANVAY CORE — one Spring Boot deployable · modules enforced at build (Modulith + ArchUnit) · payload never persisted

CONTROL PLANE — decides

identity citizen links · no auto-merge
consent purpose · revoke · issues the grant
catalog departments · schemas · journeys
registry pointers only — what exists where



60s grant
single-use

DATA PLANE — moves

orchestration BPMN journey · retry / degrade
connector adapters + mapping DSL
tracking one reference number
notifications events · citizen / officer

CROSS-CUTTING AUDIT — hash-chained ledger

Every grant, fetch, retry, consent revoke. Tamper a row → chain verify fails. Live fetch with provenance; certificates stay at the issuer.

Issuers stay systems of record — fetch on demand

Aaple Sarkar

SOAP / REST

Mahabhulekh

7/12 land

Fire e-approval

NOC API

MPCB / MAITRI

JDBC / REST

MahaDBT

REST + OAuth

DigiLocker*

consent pipe

Path: identity link + consent → signed grant → adapter fetch (not stored) → canonical map → track / officer. *DigiLocker carries issued copies; it is not a department login. Demo uses labelled sandbox + mock adapters.

Feasible now

- Core already runs: Java 21 / Boot 4 / Postgres / three journeys.
- Heterogeneous mocks are the proof (SOAP, REST, JDBC, SFTP path).
- PS constraint met: no replacement of existing department systems.
- Onboarding a journey is catalog + connectors, not a rewrite.
- Demo is fully offline — venue network is not a dependency.

Challenges / risks

- Live DigiLocker / Aadhaar needs a MeitY partner account (not on this laptop).
- Real department APIs need MoU / NIC / department credentials.
- Keycloak brokering and BPMN depth are fiddly in production.
- Probabilistic identity merge is unsafe if auto-applied.
- Officer queues and SLA if a source stays down.

How we overcome them

- Sandbox labelled as sandbox; issuer mocks stand in for SoR APIs.
- WorkflowEngine port: BPMN today, YAML DAG fallback in ~2 days.
- CHECK + service + RBAC: probabilistic match never becomes an active link.
- Degraded mode + officer retry (Revenue unavailable in demo).
- Recorded backup walkthrough if live demo fails.

IMPACT AND BENEFITS

Citizens

One issued income / caste / 7/12 / marksheet reused across services. One reference number. Consent that can be revoked — next fetch denied; nothing to purge.

Officers

One file, not three portals. See which issuer answered. Retry a down department without restarting the citizen journey.

Departments

Keep their system of record. Plug in via adapter + mapping. A new scheme is configuration (farmer subsidy already demonstrates this).

State / economic

Avoid a fourth silo or a central data lake. Interoperability layer is cheaper to extend than $N \times N$ bilateral integrations.

Social / privacy

Sensitive discovery (e.g. caste pointer) is consent-gated. Provenance on every field. Tamper-evident audit for RTI / inquiry.

Environment

Fewer duplicate physical certificates and counter visits; digital reuse of already-issued records.

Problem & policy

- SIH26129 — GoM / MSINS: interoperability without replacing existing systems
- India Stack / DEPA-style consent artefacts (purpose, requester, revoke)
- MeitY DigiLocker issued-document model (issuer push + citizen pull)

Maharashtra systems of record (not scraped; adapted)

- Aaple Sarkar — aaplesarkar.mahaonline.gov.in — income / caste
- Mahabhulekh — bhulekh.mahabhumi.gov.in — 7/12 Record of Rights
- MahaDBT — mahadbt.maharashtra.gov.in — scholarship / farmer DBT
- Fire e-approval — mahafireservice.gov.in/e-fire.php
- MAITRI / MPCB — maitri.maharashtra.gov.in — pollution consent
- MahaVastu / BPMS — mahavastu.maharashtra.gov.in

Standards we implement against

- BPMN 2.0 for cross-department journeys
- OIDC / SAML via Keycloak identity brokering
- REST, SOAP/XML, SFTP/CSV, JDBC (legacy without an API)
- JSON Schema canonical model + JSONPath mapping DSL
- Hash-chained audit (tamper-evident ledger)

This prototype (design + code)

- docs/architecture/HLD.md and LLD.md in the Samanvay repo
- docs/demo/ONE_PAGER.md · JUDGE_SCRIPT.md
- github.com/himanshu2394i/samanvay

Honest demo boundary: issuer APIs and DigiLocker partner credentials

are mocked / sandboxed until departments and MeitY onboard.